Yujing Wu | Curriculum Vitae

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Academic and Research Positions

- **Postdoctoral Researcher**
  LTE, CNRS, Paris Observatory, PSL Université, Sorbonne Université, Paris, France, Mar 2024 – present

Education

- **Doctor of Natural Science**
  School of Earth and Space Sciences, Peking University, Beijing, China, Sept 2019 – Jan 2024
  Major: Structural Geology
- **Visiting Scholar**
  Lamont-Doherty Earth Observatory, Columbia University, New York, USA, May 2022 – Sept 2023
  Major: Cyclostratigraphy
- **Bachelor of Science**
  School of Earth and Space Sciences, Peking University, Beijing, China, Sept 2015 – Jul 2019
  Major: Geochemistry
- **Bachelor of Economics**
  National School of Development, Peking University, Beijing, China, Sept 2016 – Jul 2019
  Major: Economics

Research Interests

Data-driven Earth system science; Earth–Moon system evolution; astronomical forcing of Earth system processes; cyclostratigraphy and astrochronology; zircon U–Pb geochronology database; crustal evolution; Bayesian inference; geological time-series analysis; geological big data.

Publications

In Press

1. Wang, M., Shi ,Y., Yang, A., Zhang, H., Zhang, X., **Wu Y.**, Shi, Y., Liu, J., Wei, R., & Jin, Z., Coupled sea-level and biodiversity dynamics during the Cambrian SPICE event. *Nature Communications*. In press.

Peer-reviewed Publications

9. Hoang, N., Laskar, J., Hara, N. C., **Wu, Y.**, Sultanov, A., Sinnesael, M., Westerhold, T., & Bujons, P. (2025). AstroGeoFit. A genetic algorithm and Bayesian approach for the astronomical calibration of the geological timescale. *Paleoceanography and Paleoclimatology*, *40*(8), e2024PA005021. https://doi.org/10.1029/2024PA005021
8. **Wu, Y.**, Malinverno, A., Meyers, S. R., & Hinnov, L. A. (2024). A 650-Myr history of Earth's axial precession frequency and the evolution of the Earth-Moon system derived from cyclostratigraphy. *Science Advances*, *10*(42), eado2412. https://doi.org/10.1126/sciadv.ado2412
7. Wang, M., Li, M., Hajek, E. A., Kemp, D. B., **Wu, Y.**, Zhu, H., Huang, C., Zhang, H., Ji, K., Zhang, R., Wei, R., & Jin, Z. (2024). Assessing the preservation of orbital signals across different sedimentary environments: Insights from stochastic sedimentation modeling. *Earth and Planetary Science Letters*, *642*, 118866. https://doi.org/10.1016/j.epsl.2024.118866
6. **Wu, Y.**, Fang, X., & Ji, J., (2023). A global zircon U–Th–Pb geochronological database. *Earth System Science Data*, *15*(11), 5171-5181. https://doi.org/10.5194/essd-15-5171-2023
5. **Wu, Y.**, Fang, X., Jiang, L., Song, B., Han, B., Li, M., & Ji, J. (2022). Very long-term periodicity of episodic zircon production and Earth system evolution. *Earth-Science Reviews*, 104164. https://doi.org/10.1016/j.earscirev.2022.104164
4. **Wu, Y.**, Fang, X., Liao, S., Xue, L., Chen, Z., Yang, J., Lu, Y., Ling, K., Hu, S., Kong, S., Xiong, Y., Li, H., Shang, X., Ji, R., Lu, X., Song, B., Zhang, L., & Ji, J. (2020). Crustal evolution events in the Chinese continent: evidence from a zircon U-Pb database. *International Journal of Digital Earth*, *13*(12), 1532-1552. https://doi.org/10.1080/17538947.2020.1739152
3. Fang, X., **Wu, Y.**, Liao, S., Xue, L., Chen, Z., Yang, J., Lu, Y., Ling, K., Hu, S., Kong, S., Xiong, Y., Li, H., Shang, X., Ji, R., Lu, X., Song, B., Zhang, L., & Ji, J. (2020). Division of crustal units in China using grid-based clustering and a zircon U–Pb geochronology database. *Computers & Geosciences*, *145*, 104570. https://doi.org/10.1016/j.cageo.2020.104570
2. **Wu, Y.**, Fang, X., Liao, S., Xue, L., Chen, Z., Yang, J., Lu, Y., Ling, K., Hu, S., Kong, S., Xiong, Y., Li, H., Shang, X., Ji, R., Lu, X., Song, B., Zhang, L., & Ji, J. (2019). Zircon U-Pb geochronology of the Chinese continental crust: a preliminary analysis of the Elsevier science database. *Big Earth Data*, 3(1), 26-44. https://doi.org/10.1080/20964471.2019.1576261
1. Fang, X., **Wu, Y.**, Liao, S., Xue, L., Chen, Z., Song, B., Zhang, L., & Ji, J. (2018). Preliminary analysis of the Chinese sub-database of the Chinese crust single-grain zircon U-Pb geochronology database. *ACTA PETROLOGICA SINICA*, *34*(11), 3253-3265 (in Chinese).

Manuscripts in Preparation

2. **Wu, Y.**, Laskar, J., Westerhold, T., Thibault, N., Sultanov, A., Hara, N., Bujons, P., & Gastineau, M. AstroGeoFit-based reconstruction of astronomical time scales and orbital eccentricity over 40–100 Ma from 47 cyclostratigraphic records.
1. **Wu, Y.**, Fang, X., & Ji, J. Precise age constraints on the peak periods of the cyclical assembly and breakup of supercontinents.

Research Databases, Software, and Open Science Contributions

Public Research Databases

Wu, Y., Fang, X., & Ji, J. (2023). Global zircon U-Th-Pb geochronology database (Version 2.1) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.7387566. >3,000 downloads on Zenodo.

Open-Source Python Package

AstroGeoFit — Contributed to code development, validation, and application of a genetic-algorithm and Bayesian framework for astronomical calibration of geological time series. https://astrogeo.eu/astrogeofit/

Fellowships and Awards (Selected)

- Excellent Graduate (doctoral level), Beijing Municipal Education Commission, 2024
- President Fellowship, Peking University, 2022– 2023
- China Aerospace Science and Technology Corporation Scholarship, Peking University, 2021– 2022
- Sun Hung Kai Properties Scholarship, Peking University, 2019
- Excellent Graduate (undergraduate level), Beijing Municipal Education Commission, 2019
- Langrun Youth Scholarship, 2017 – 2018
- Outstanding Field Geology Report Award, Peking University, 2019
- Cyrus Tang Scholarship, Peking University, 2015 – 2019

Invited Talks and Conference Presentations (Selected)

Invited Talk

- Invited research seminar, Peking University, Sept 2025.
  Title: A 650-Myr history of Earth's axial precession frequency and the evolution of the Earth-Moon system derived from cyclostratigraphy

Conference Presentations

6. **<u>Wu, Y.</u>**, Laskar, J., Westerhold T., et al., (2026).
   AstroGeoFit in action. Towards an Eccentricity-Based Astronomical Time Scale for 40–100 Ma.
   GSA 2026 Penrose Conference, Iowa City IA, USA, May 30 – Jun 05.
5. **<u>Wu, Y.</u>**, Laskar, J., Westerhold T., et al., (2026).
   AstroGeoFit in action. Towards an Eccentricity-Based Astronomical Time Scale for 40–100 Ma.
   EGU General Assembly 2026, Vienna, Austria, May 04–08.
4. **<u>Wu, Y.</u>**, Malinverno, A., Meyers, S. R., et al., (2023).
   Earth's axial precession frequency, length of day, lunar distance, and insolation periodicities over the past 650 Myr from cyclostratigraphy.
   AGU Fall Meeting 2023, San Francisco CA, USA, Dec 11-15.
3. **<u>Wu, Y.</u>,** Fang, X., & Ji, J., (2023).
   Implications of zircon Th/U for global continental crustal evolution and geodynamics.
   EGU General Assembly 2023, Vienna, Austria, Apr 24–28.
2. **<u>Wu, Y.</u>,** Fang, X., Jiang, L., et al., (2022).
   Very long-term periodicity of episodic zircon production and Earth system evolution.
   AGU Fall Meeting 2022, Chicago IL, USA, Dec 12 –16.
1. **<u>Wu, Y.</u>**, Fang, X., Liao, S., et al., (2018).
   Great events of Chinese continental evolution: implications from zircon U-Pb geochronology database.
   AGU Fall Meeting 2018, Washington DC, USA, Dec 10 –14.

Teaching Experience

Teaching Assistant, *Introduction to Earth Sciences*, undergraduate course, Peking University, Sept. 2020–Jan. 2021
Led tutorial and practical sessions; organized Q&A sessions; supervised a geology field trip; graded assignments and contributed to student assessment; occasionally substituted for the instructor in lectures. Approx. 60 teaching hours.

Student Mentorship

Xiaoyu Zhang – Visiting Ph.D. student, Paris Observatory, 2026 – Present

Project: Reconstruction of the astronomical time scale of the Mesoproterozoic.

Can Cai – Visiting Ph.D. student, Paris Observatory, 2026 – Present

Project: Reconstruction of climatic precession evolution during the Paleogene.

Muyuan Zhu – Undergraduate student, Peking University, 2021 – 2022

Project: Compilation and analysis of zircon U–Pb geochronology databases.

Professional Service, Workshops, and Outreach

- **Journal reviewer** for *GSA Bulletin*, *Scientific Reports*, and *Big Earth Data*.
- **Workshop co-host**, AstroGeoFit International Workshop, Paris, France, May 25–29, 2026. Contributed to the organization of the workshop and led hands-on practical sessions using the AstroGeoFit open-source software.
- **Session co-convener and co-chair**, "Fall Astronomical Forcing of Earth's Paleoclimate System" (PP53G), AGU Fall Meeting, Washington, DC, USA, December 2024.

Skills

- **Computational & Data Analysis Skills:**
  - Large-scale geoscientific data integration and database management
  - Time series analysis, signal detection, and spectral methods across multi-scale geological records
  - Bayesian statistics and probabilistic modeling
  - Machine learning and artificial intelligence methods for geoscientific data analysis, with Python-based training
- **Programming Languages:** Python, MATLAB, R, SQL, Shell scripting
- **Languages:** Chinese (native), English (C1), French (B2)